Project 3: Pedestrian Detection in Urban Environments — Exploratory Data Analysis

We Came, We Saw, We Modelled | Emily Friedman (Group Leader), Jensen Harvey, Kaitlyn Chou

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```r
knitr::opts_chunk$set(
  echo      = TRUE,
  warning   = FALSE,
  message   = FALSE,
  fig.align = "center",
  fig.width  = 7,
  fig.height = 4.5
)
```

```r
library(jsonlite)
library(tidyverse)
library(ggplot2)
library(scales)
library(patchwork)
```

---

1 Data Loading

```r
data_dir <- "/Users/jensen.harvey/Desktop/data science/project3/data"

json_files <- list.files(
  path      = data_dir,
  pattern   = "gtBboxCityPersons\\.json$",
  recursive = TRUE,
  full.names = TRUE
)

cat("Total annotation files found:", length(json_files), "\n")
```

```
## Total annotation files found: 3475
```

```r
`%||%` <- function(a, b) if (!is.null(a)) a else b

parse_bbox_file <- function(filepath) {
  parts <- strsplit(filepath, "/")[[1]]
  n     <- length(parts)
  city  <- parts[n - 1]
  split <- parts[n - 2]

  ann <- tryCatch(fromJSON(filepath, simplifyVector = FALSE), error = function(e) NULL)
  if (is.null(ann)) return(NULL)

  img_w <- ann$imgWidth  %||% NA_integer_
  img_h <- ann$imgHeight %||% NA_integer_
```

```r
  objects <- ann$objects
  if (is.null(objects) || length(objects) == 0) {
    return(data.frame(
      file       = filepath,
      split      = split,
      city       = city,
      img_width  = img_w,
      img_height = img_h,
      label      = NA_character_,
      bbox_x     = NA_real_,
      bbox_y     = NA_real_,
      bbox_w     = NA_real_,
      bbox_h     = NA_real_,
      bbox_vis_w = NA_real_,
      bbox_vis_h = NA_real_,
      stringsAsFactors = FALSE
    ))
  }

  rows <- lapply(objects, function(o) {
    bbox     <- o$bbox    %||% list(0, 0, 0, 0)
    bbox_vis <- o$bboxVis %||% list(0, 0, 0, 0)
    data.frame(
      file       = filepath,
      split      = split,
      city       = city,
      img_width  = img_w,
      img_height = img_h,
      label      = o$label %||% NA_character_,
      bbox_x     = bbox[[1]],
      bbox_y     = bbox[[2]],
      bbox_w     = bbox[[3]],
      bbox_h     = bbox[[4]],
      bbox_vis_w = bbox_vis[[3]],
      bbox_vis_h = bbox_vis[[4]],
      stringsAsFactors = FALSE
    )
  })
  do.call(rbind, rows)
}

annotations <- map_dfr(json_files, parse_bbox_file)

cat("Total annotated object rows:", nrow(annotations), "\n")
```

```
## Total annotated object rows: 33775
```

```r
cat("Unique images:", n_distinct(annotations$file), "\n")
```

```
## Unique images: 3475
```

```r
cat("Splits:", paste(sort(unique(annotations$split)), collapse = ", "), "\n")
```

```
## Splits: train, val
```

```r
cat("Cities:", paste(sort(unique(annotations$city)), collapse = ", "), "\n")
```

```
## Cities: aachen, bochum, bremen, cologne, darmstadt, dusseldorf, erfurt, frankfurt, hamburg, hanover,
```

---

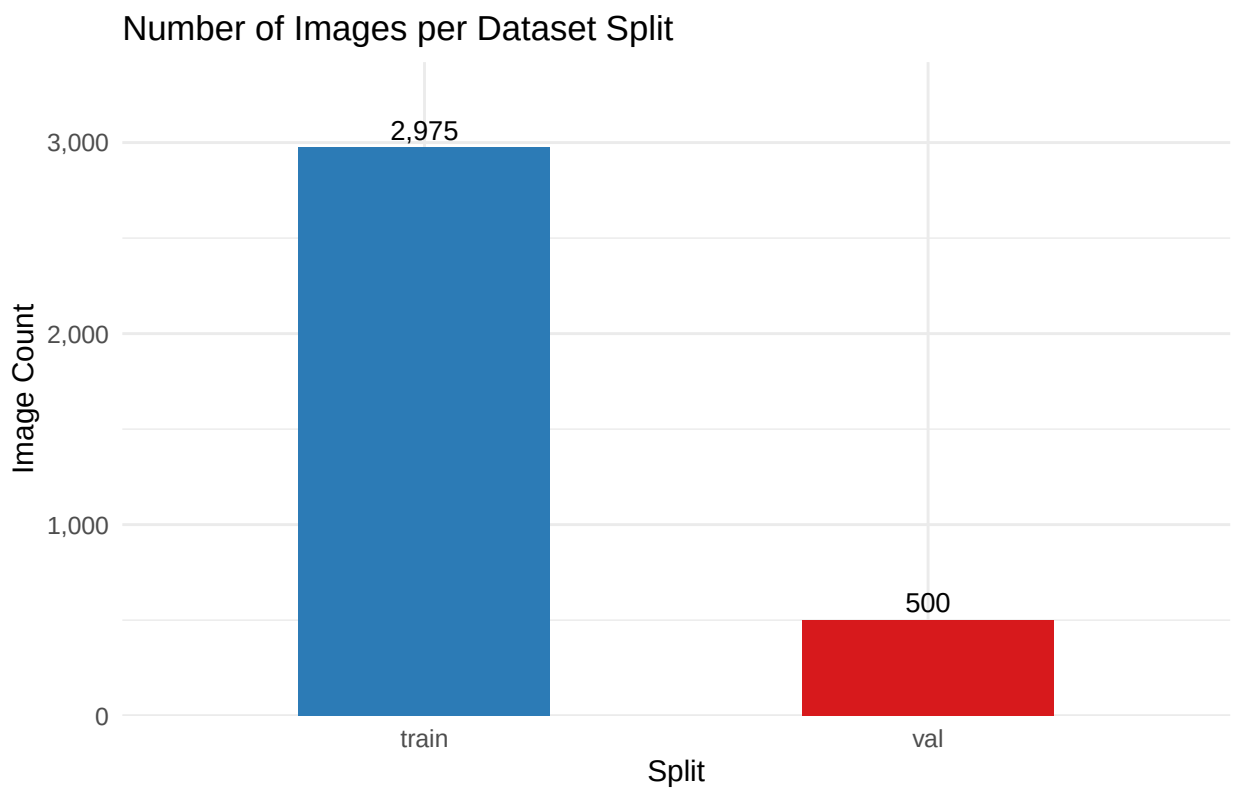
2 Dataset Overview

2.1 Image Count by Split

```r
images_df <- annotations %>%
  distinct(file, split, city, img_width, img_height)

split_counts <- images_df %>%
  count(split, name = "n_images") %>%
  arrange(desc(n_images))

split_counts
```

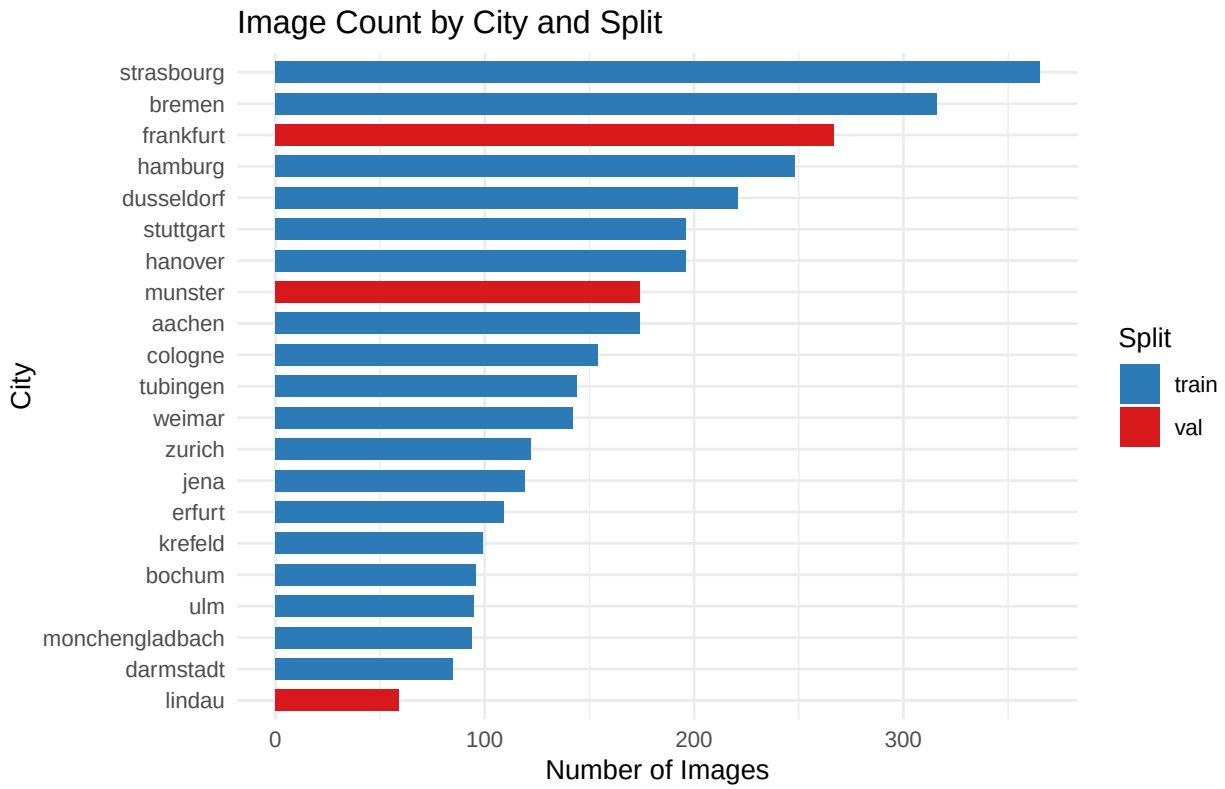
```
##   split n_images
## 1 train     2975
## 2   val      500
```

```r
ggplot(split_counts, aes(x = reorder(split, -n_images), y = n_images, fill = split)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = comma(n_images)), vjust = -0.4, size = 4) +
  scale_fill_manual(values = c("train" = "#2c7bb6", "val" = "#d7191c")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15)), labels = comma) +
  labs(
    title = "Number of Images per Dataset Split",
    x     = "Split",
    y     = "Image Count"
  ) +
  theme_minimal(base_size = 12)
```



2.2 Image Count by City

```
city_counts <- images_df %>%
  count(city, split, name = "n_images")

ggplot(city_counts, aes(x = reorder(city, n_images), y = n_images, fill = split)) +
  geom_col(width = 0.7) +
  coord_flip() +
  scale_fill_manual(values = c("train" = "#2c7bb6", "val" = "#d7191c")) +
  labs(
    title = "Image Count by City and Split",
    x     = "City",
    y     = "Number of Images",
    fill  = "Split"
  ) +
  theme_minimal(base_size = 11)
```



---

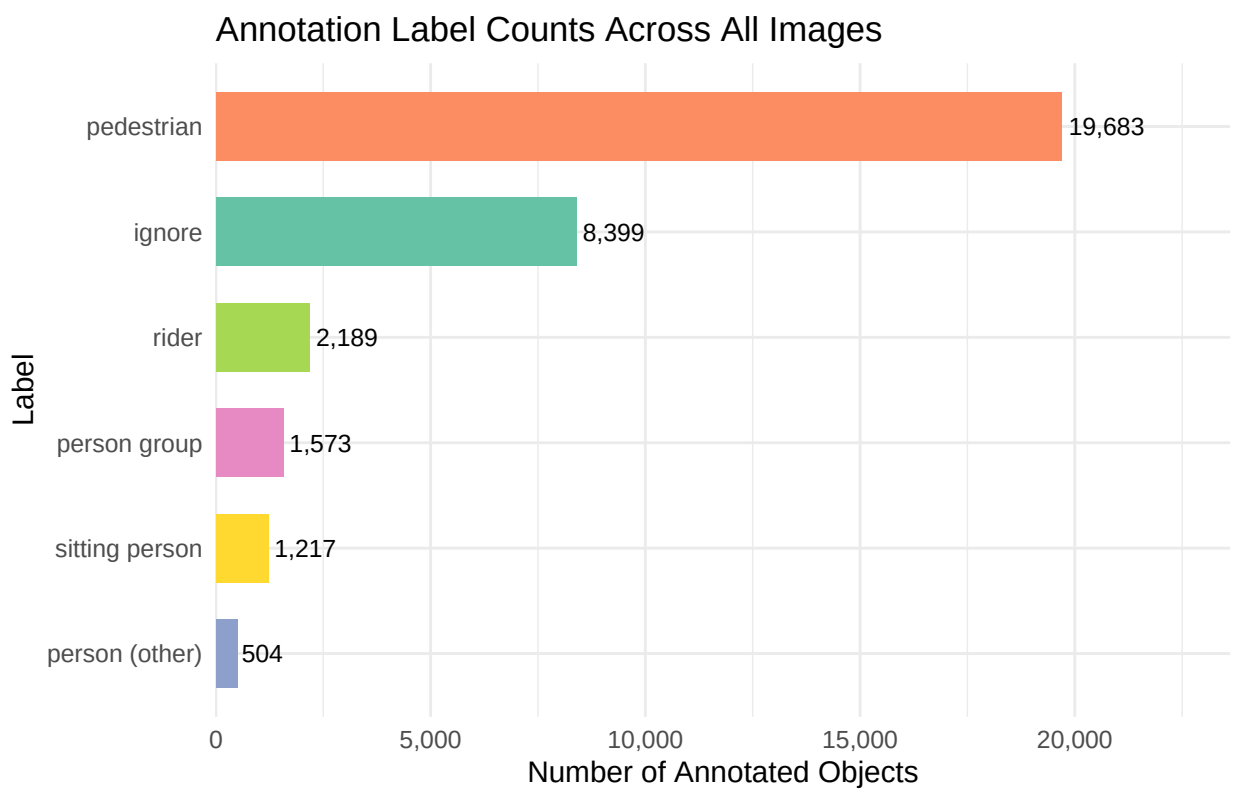
3 Label Distribution

```
label_counts <- annotations %>%
  filter(!is.na(label)) %>%
  count(label, name = "n_objects") %>%
  arrange(desc(n_objects))

label_counts
```

```
##            label n_objects
## 1     pedestrian     19683
## 2         ignore      8399
## 3          rider      2189
## 4   person group      1573
## 5 sitting person      1217
## 6 person (other)       504
```

```
ggplot(label_counts, aes(x = reorder(label, n_objects), y = n_objects, fill = label)) +
  geom_col(width = 0.65, show.legend = FALSE) +
  geom_text(aes(label = comma(n_objects)), hjust = -0.1, size = 3.5) +
  coord_flip() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.2)), labels = comma) +
```

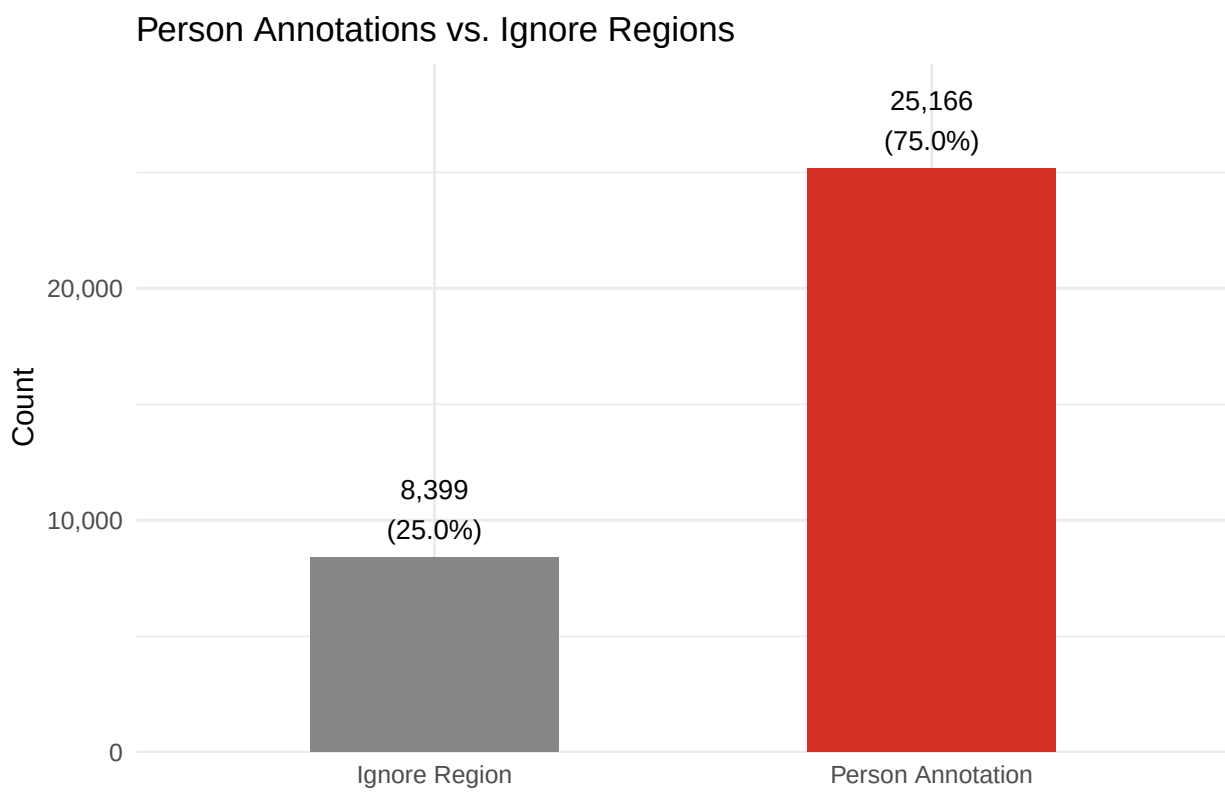
```r
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Annotation Label Counts Across All Images",
    x     = "Label",
    y     = "Number of Annotated Objects"
  ) +
  theme_minimal(base_size = 12)
```



3.1 Person Annotations vs. Ignore Regions

```r
annotations <- annotations %>%
  mutate(
    is_ignore = grepl("ignore", label, ignore.case = TRUE),
    is_person = !is_ignore & !is.na(label)
  )

type_summary <- annotations %>%
  filter(!is.na(label)) %>%
  count(is_ignore) %>%
  mutate(
    type = ifelse(is_ignore, "Ignore Region", "Person Annotation"),
    pct  = n / sum(n)
  )
```

```r
ggplot(type_summary, aes(x = type, y = n, fill = type)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = paste0(comma(n), "\n(", percent(pct, accuracy = 0.1), ")")),
            vjust = -0.3, size = 4) +
  scale_fill_manual(values = c("Person Annotation" = "#d73027",
                               "Ignore Region"     = "#878787")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.18)), labels = comma) +
  labs(
    title = "Person Annotations vs. Ignore Regions",
    x     = NULL, y = "Count"
  ) +
  theme_minimal(base_size = 12)
```



---

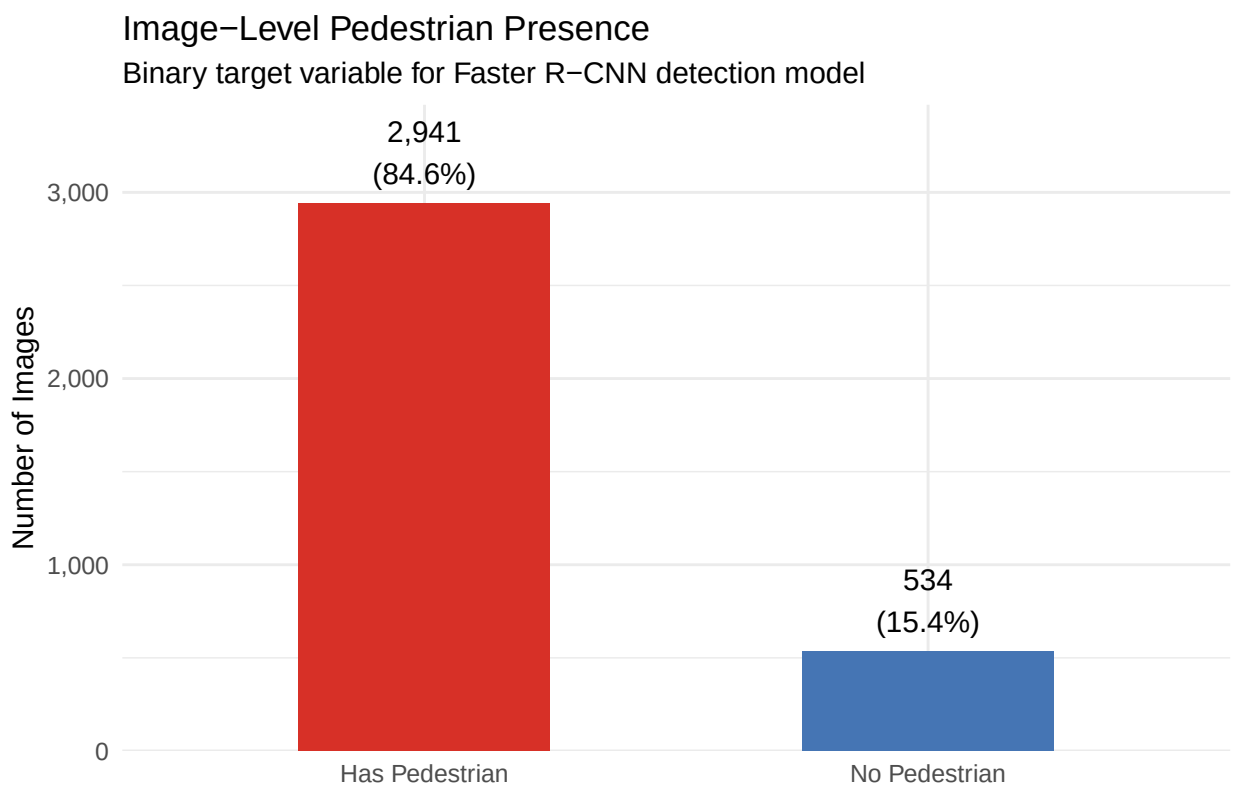
4 Pedestrian Presence per Image (Binary Target Variable)

```r
image_level <- annotations %>%
  group_by(file, split, city, img_width, img_height) %>%
  summarise(
    n_persons        = sum(is_person,  na.rm = TRUE),
    n_ignore_regions = sum(is_ignore,  na.rm = TRUE),
    has_pedestrian   = any(is_person,  na.rm = TRUE),
```

```r
    .groups = "drop"
  )

presence_summary <- image_level %>%
  count(has_pedestrian) %>%
  mutate(
    label = ifelse(has_pedestrian, "Has Pedestrian", "No Pedestrian"),
    pct   = n / sum(n)
  )

presence_summary
```

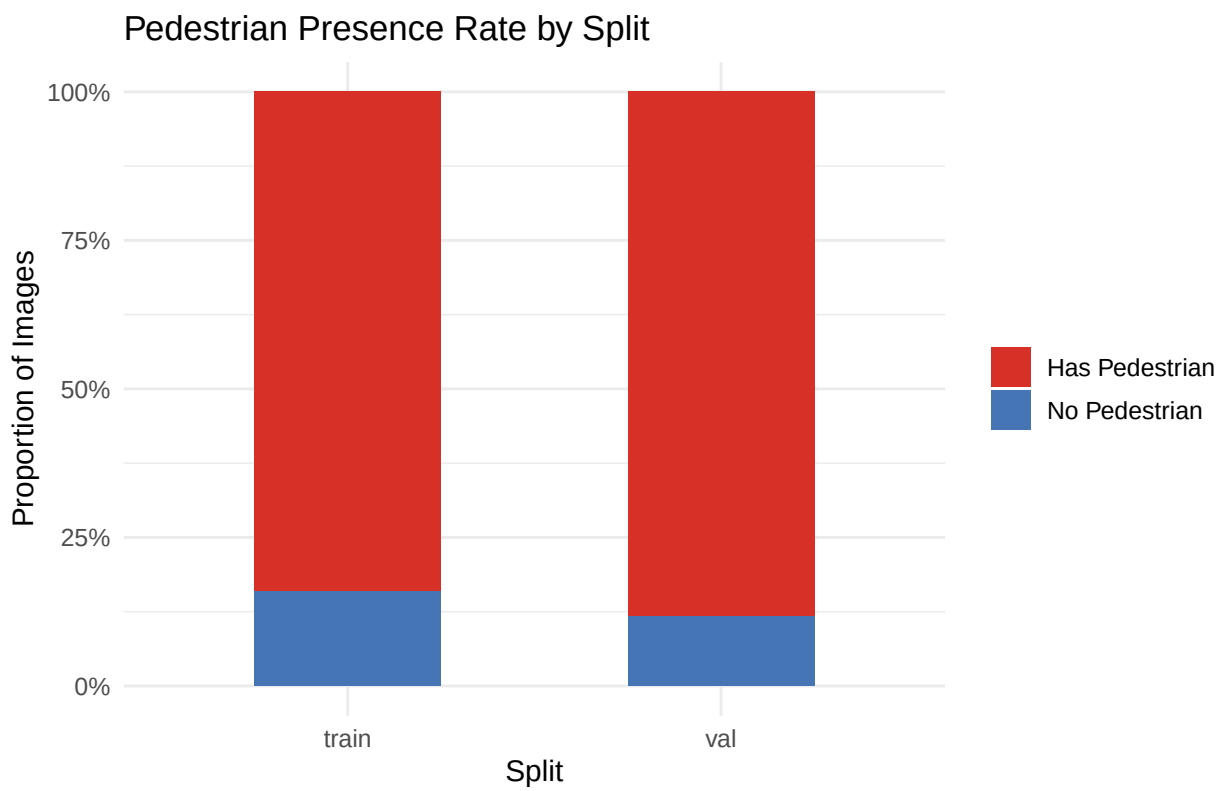
```
## # A tibble: 2 x 4
##   has_pedestrian     n label            pct
##   <lgl>          <int> <chr>          <dbl>
## 1 FALSE            534 No Pedestrian  0.154
## 2 TRUE            2941 Has Pedestrian 0.846
```

```r
ggplot(presence_summary, aes(x = label, y = n, fill = label)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = paste0(comma(n), "\n(", percent(pct, accuracy = 0.1), ")")),
            vjust = -0.3, size = 4.2) +
  scale_fill_manual(values = c("Has Pedestrian" = "#d73027",
                               "No Pedestrian"  = "#4575b4")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.18)), labels = comma) +
  labs(
    title    = "Image-Level Pedestrian Presence",
    subtitle = "Binary target variable for Faster R-CNN detection model",
    x = NULL, y = "Number of Images"
  ) +
  theme_minimal(base_size = 12)
```



4.1 Presence Rate by Split

```
split_presence <- image_level %>%
  group_by(split, has_pedestrian) %>%
  summarise(n = n(), .groups = "drop") %>%
  mutate(label = ifelse(has_pedestrian, "Has Pedestrian", "No Pedestrian"))

ggplot(split_presence, aes(x = split, y = n, fill = label)) +
  geom_col(position = "fill", width = 0.5) +
  scale_y_continuous(labels = percent) +
  scale_fill_manual(values = c("Has Pedestrian" = "#d73027",
                               "No Pedestrian"  = "#4575b4")) +
  labs(
    title = "Pedestrian Presence Rate by Split",
    x = "Split", y = "Proportion of Images", fill = NULL
  ) +
  theme_minimal(base_size = 12)
```

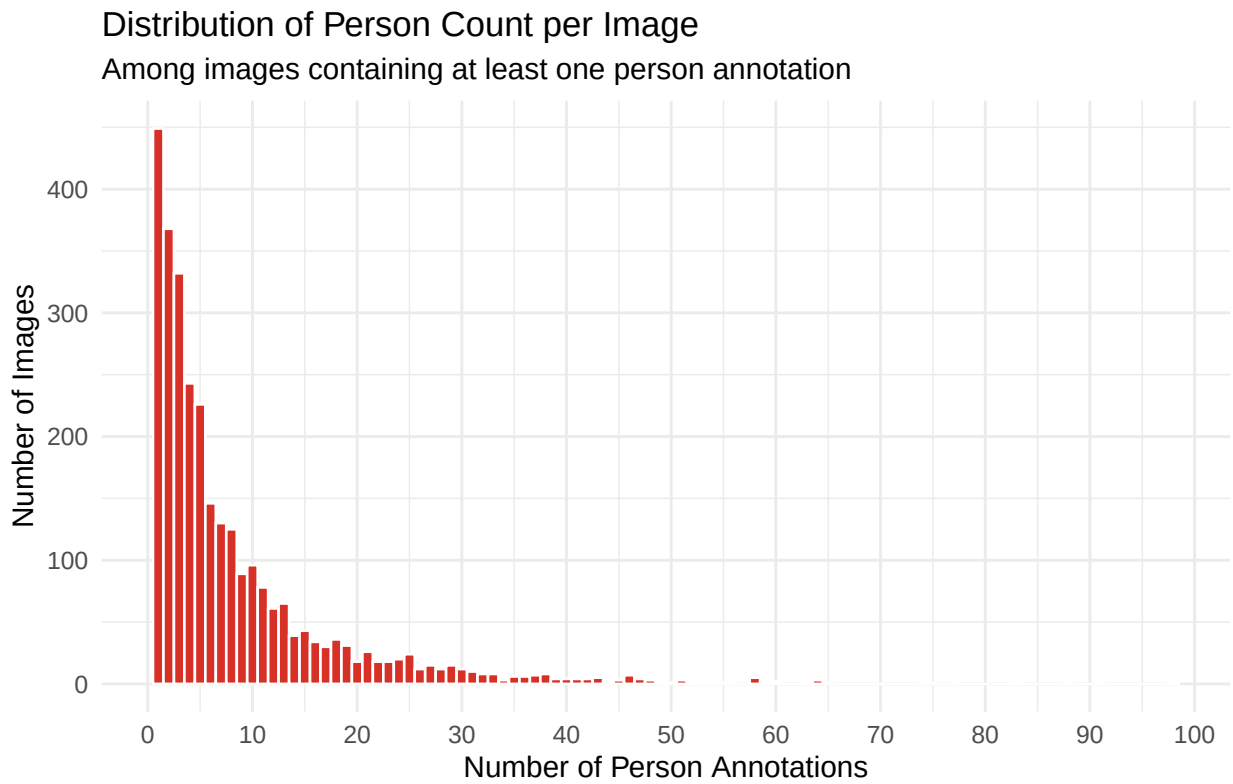


---

5 Pedestrian Count per Image

5.1 Histogram

```
image_level %>%
  filter(has_pedestrian) %>%
  ggplot(aes(x = n_persons)) +
  geom_histogram(binwidth = 1, fill = "#d73027", color = "white") +
  scale_x_continuous(breaks = pretty_breaks(n = 10)) +
  labs(
    title    = "Distribution of Person Count per Image",
    subtitle = "Among images containing at least one person annotation",
    x = "Number of Person Annotations", y = "Number of Images"
  ) +
  theme_minimal(base_size = 12)
```



5.2 Summary Statistics

```
image_level %>%
  summarise(
    total_images           = n(),
    images_with_pedestrian = sum(has_pedestrian),
    images_without         = sum(!has_pedestrian),
    pct_with_pedestrian    = mean(has_pedestrian),
    mean_persons_all       = mean(n_persons),
    mean_persons_ped_only  = mean(n_persons[has_pedestrian]),
    median_persons_ped     = median(n_persons[has_pedestrian]),
    max_persons            = max(n_persons)
  ) %>%
  pivot_longer(everything(), names_to = "Metric", values_to = "Value") %>%
  mutate(Value = round(Value, 3))
```

```
## # A tibble: 8 x 2
##   Metric                    Value
##   <chr>                     <dbl>
## 1 total_images           3475
## 2 images_with_pedestrian 2941
## 3 images_without          534
## 4 pct_with_pedestrian       0.846
## 5 mean_persons_all          7.24
## 6 mean_persons_ped_only     8.56
```

```
## 7 median_persons_ped        5
## 8 max_persons              98
```

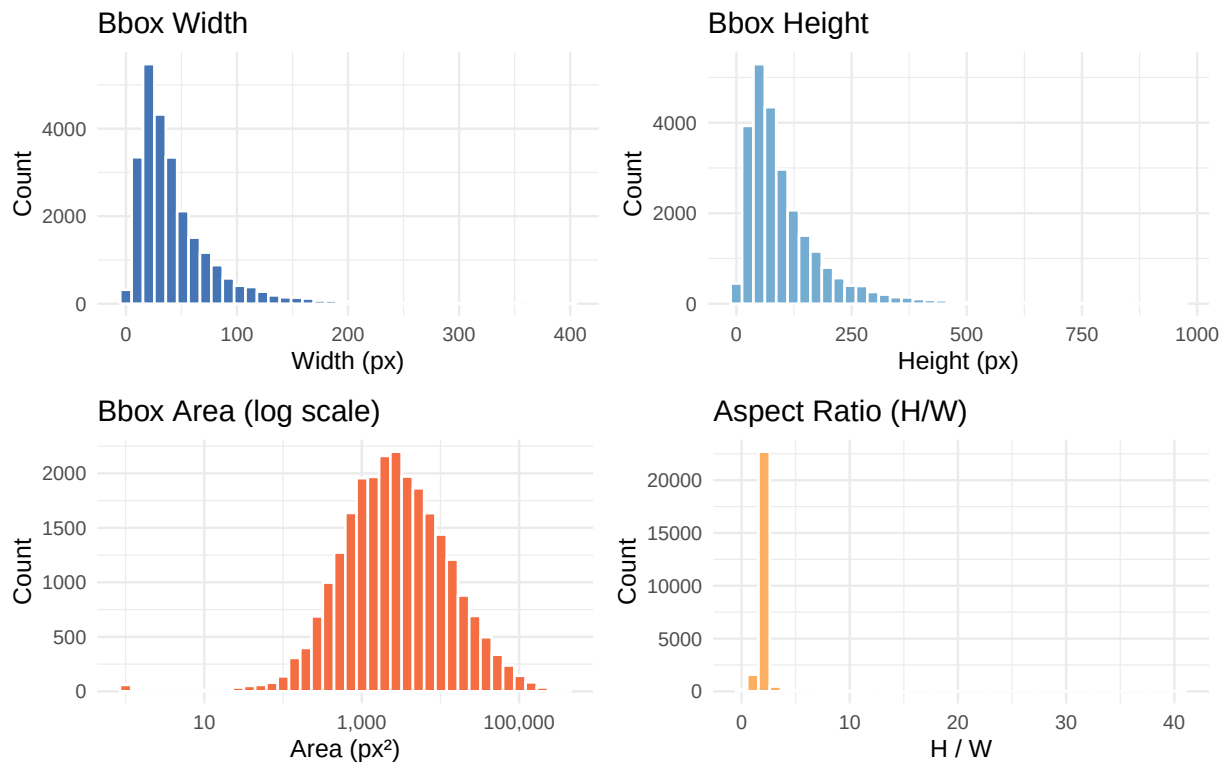
---

6 Bounding Box Analysis

```r
bbox_df <- annotations %>%
  filter(is_person, !is.na(bbox_w), !is.na(bbox_h), bbox_w > 0, bbox_h > 0) %>%
  mutate(
    bbox_area    = bbox_w * bbox_h,
    aspect_ratio = bbox_h / bbox_w,
    vis_area     = bbox_vis_w * bbox_vis_h,
    visibility   = ifelse(bbox_area > 0, vis_area / bbox_area, NA_real_)
  )
```

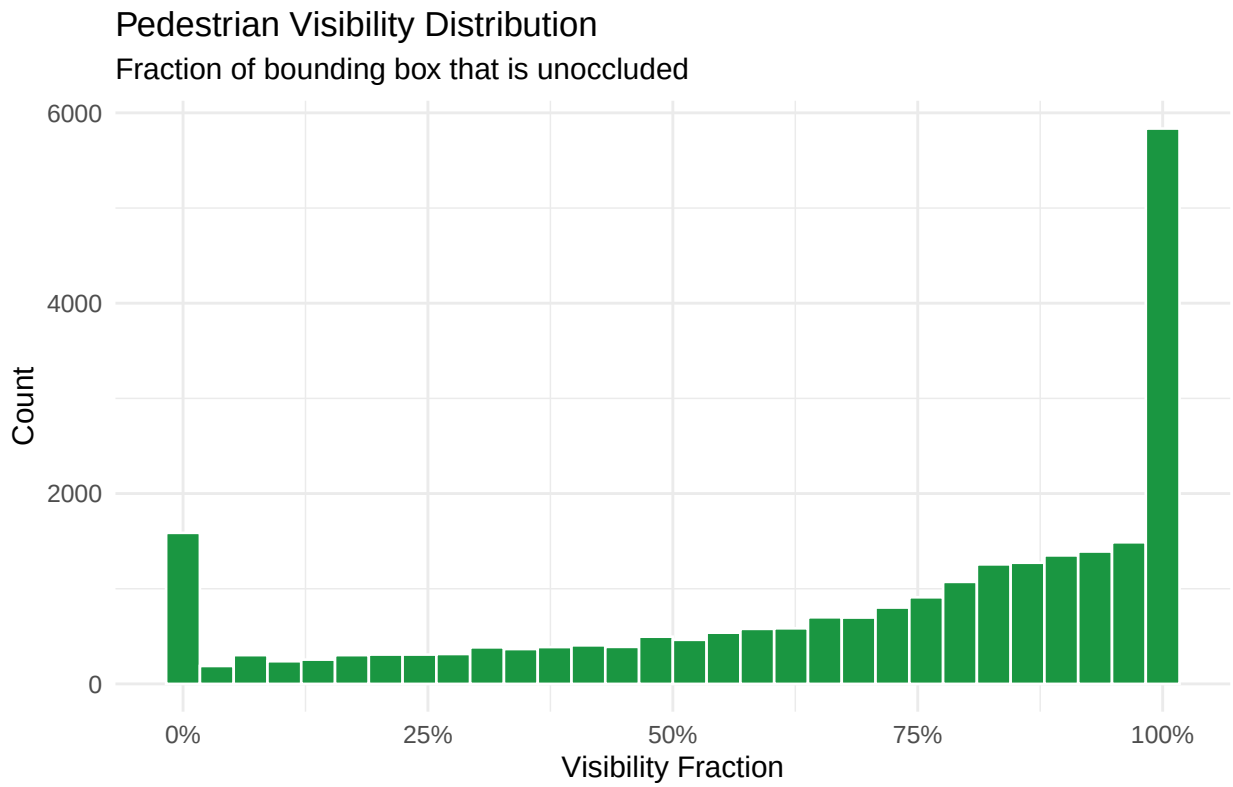
6.1 Width, Height, Area, and Aspect Ratio

```r
p1 <- ggplot(bbox_df, aes(x = bbox_w)) +
  geom_histogram(bins = 40, fill = "#4575b4", color = "white") +
  labs(title = "Bbox Width", x = "Width (px)", y = "Count") +
  theme_minimal(base_size = 10)

p2 <- ggplot(bbox_df, aes(x = bbox_h)) +
  geom_histogram(bins = 40, fill = "#74add1", color = "white") +
  labs(title = "Bbox Height", x = "Height (px)", y = "Count") +
  theme_minimal(base_size = 10)

p3 <- ggplot(bbox_df, aes(x = bbox_area)) +
  geom_histogram(bins = 40, fill = "#f46d43", color = "white") +
  scale_x_log10(labels = comma) +
  labs(title = "Bbox Area (log scale)", x = "Area (px²)", y = "Count") +
  theme_minimal(base_size = 10)

p4 <- ggplot(bbox_df, aes(x = aspect_ratio)) +
  geom_histogram(bins = 40, fill = "#fdae61", color = "white") +
  labs(title = "Aspect Ratio (H/W)", x = "H / W", y = "Count") +
  theme_minimal(base_size = 10)

(p1 + p2) / (p3 + p4)
```



6.2 Visibility (Occlusion)

```
bbox_df %>%
  filter(visibility >= 0, visibility <= 1) %>%
  ggplot(aes(x = visibility)) +
  geom_histogram(bins = 30, fill = "#1a9641", color = "white") +
  scale_x_continuous(labels = percent) +
  labs(
    title    = "Pedestrian Visibility Distribution",
    subtitle = "Fraction of bounding box that is unoccluded",
    x = "Visibility Fraction", y = "Count"
  ) +
  theme_minimal(base_size = 12)
```



6.3 Bounding Box Summary Statistics

```
bbox_df %>%
  summarise(
    n_person_annotations = n(),
    mean_width           = mean(bbox_w),
    median_width         = median(bbox_w),
    mean_height          = mean(bbox_h),
    median_height        = median(bbox_h),
    mean_area            = mean(bbox_area),
    median_area          = median(bbox_area),
    mean_aspect_ratio    = mean(aspect_ratio),
    mean_visibility      = mean(visibility, na.rm = TRUE)
  ) %>%
  pivot_longer(everything(), names_to = "Metric", values_to = "Value") %>%
  mutate(Value = round(Value, 2))
```

```
## # A tibble: 9 x 2
##   Metric                  Value
##   <chr>                   <dbl>
## 1 n_person_annotations 25166
## 2 mean_width              44.8
## 3 median_width            33
## 4 mean_height            106.
## 5 median_height           77
```

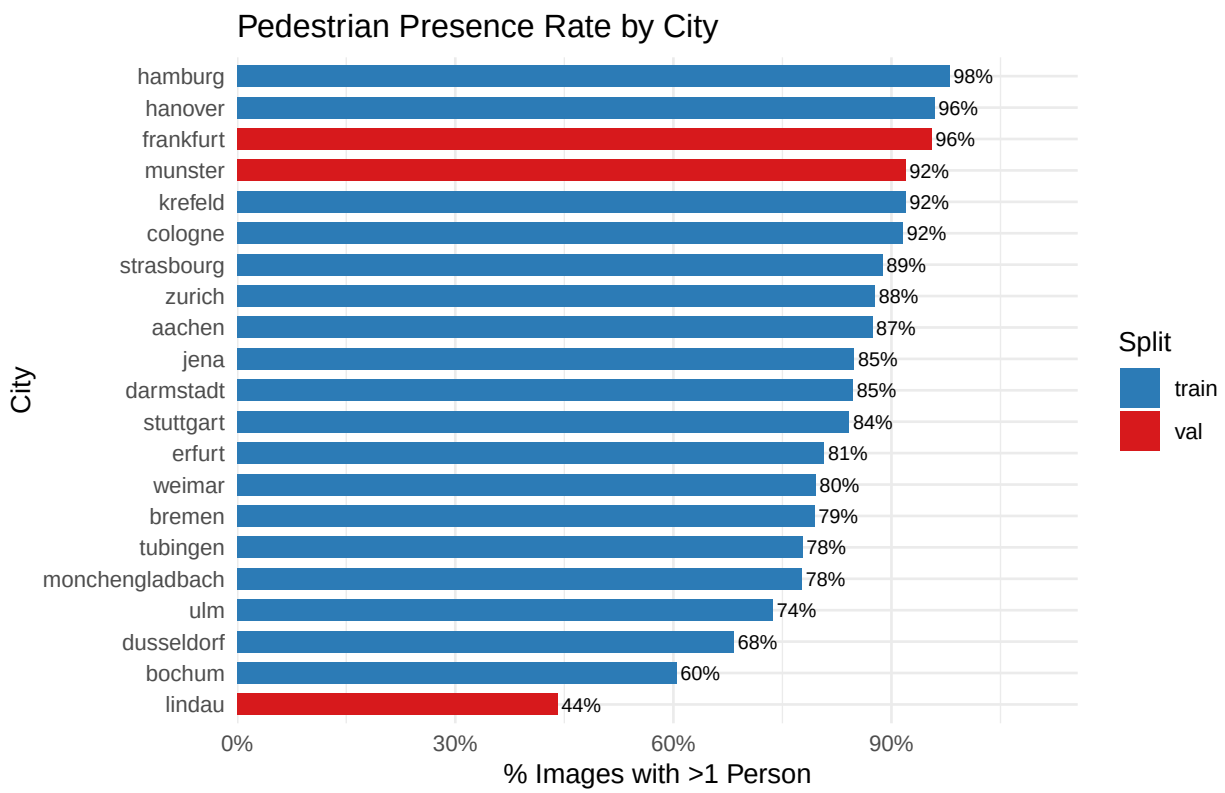
```
## 6 mean_area             8187.
## 7 median_area           2520
## 8 mean_aspect_ratio        2.37
## 9 mean_visibility          0.7
```

---

7 City-Level Analysis

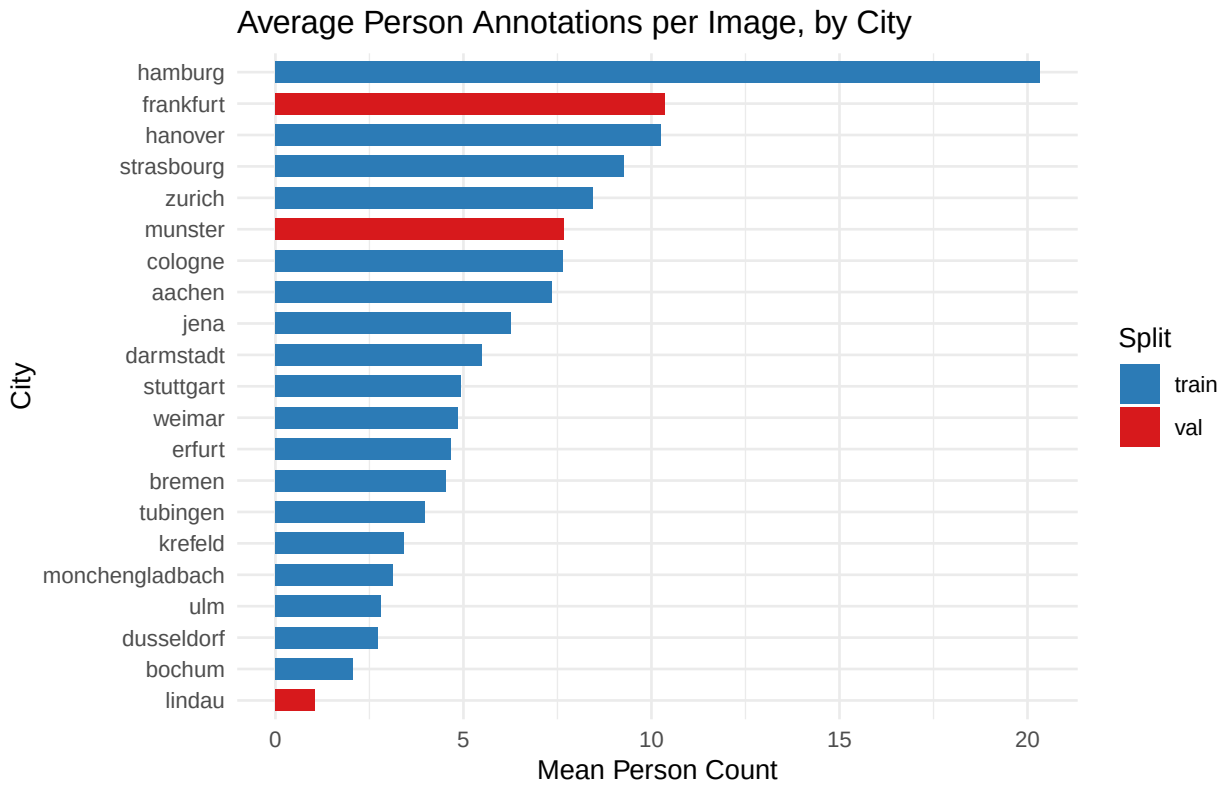
7.1 Pedestrian Presence Rate by City

```r
city_stats <- image_level %>%
  group_by(city, split) %>%
  summarise(
    n_images     = n(),
    n_with_ped   = sum(has_pedestrian),
    pct_with_ped = mean(has_pedestrian),
    avg_persons  = mean(n_persons),
    .groups      = "drop"
  )

ggplot(city_stats, aes(x = reorder(city, pct_with_ped),
                       y = pct_with_ped, fill = split)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = percent(pct_with_ped, accuracy = 1)),
            hjust = -0.1, size = 2.8) +
  coord_flip() +
  scale_y_continuous(labels = percent, expand = expansion(mult = c(0, 0.18))) +
  scale_fill_manual(values = c("train" = "#2c7bb6", "val" = "#d7191c")) +
  labs(
    title = "Pedestrian Presence Rate by City",
    x = "City", y = "% Images with >1 Person", fill = "Split"
  ) +
  theme_minimal(base_size = 11)
```



7.2 Average Person Count by City

```
ggplot(city_stats, aes(x = reorder(city, avg_persons), y = avg_persons, fill = split)) +
  geom_col(width = 0.7) +
  coord_flip() +
  scale_fill_manual(values = c("train" = "#2c7bb6", "val" = "#d7191c")) +
  labs(
    title = "Average Person Annotations per Image, by City",
    x = "City", y = "Mean Person Count", fill = "Split"
  ) +
  theme_minimal(base_size = 11)
```



---

8 Class Imbalance Assessment

```
imbalance <- image_level %>%
  count(has_pedestrian) %>%
  mutate(
    class = ifelse(has_pedestrian, "Positive (pedestrian)", "Negative (no pedestrian)"),
    pct   = n / sum(n)
  )

imbalance %>% select(class, n, pct) %>% mutate(pct = percent(pct, accuracy = 0.1))
```

```
## # A tibble: 2 x 3
##   class                        n pct
##   <chr>                    <int> <chr>
## 1 Negative (no pedestrian)   534 15.4%
## 2 Positive (pedestrian)     2941 84.6%
```

```
ratio <- max(imbalance$n) / min(imbalance$n)
cat("Imbalance ratio (majority:minority):", round(ratio, 2), "\n")
```

```
## Imbalance ratio (majority:minority): 5.51
```

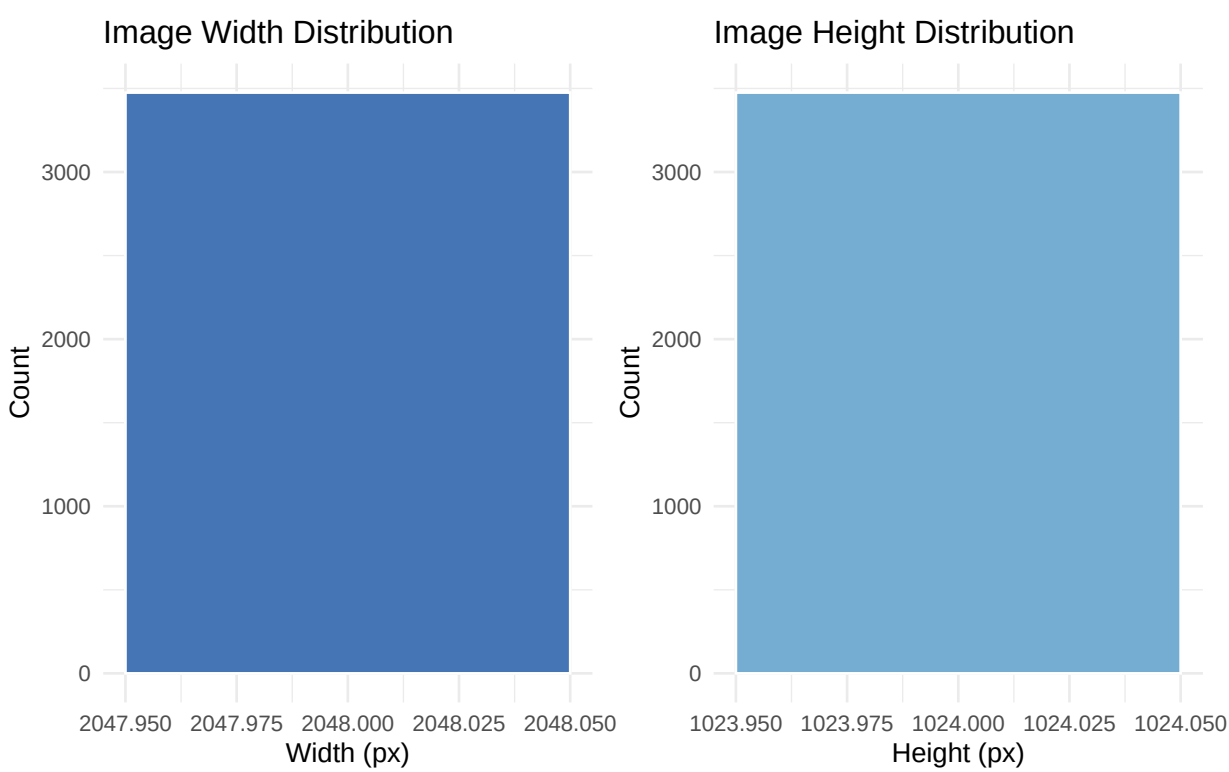
```r
cat(ifelse(ratio > 3,
  ">> Ratio > 3:1 - consider class-weighted loss or oversampling.",
  ">> Ratio <= 3:1 - class balance is acceptable for standard training."), "\n")
```

```
## >> Ratio > 3:1 - consider class-weighted loss or oversampling.
```

---

9 Image Dimensions

```r
dims_df <- images_df %>% filter(!is.na(img_width), !is.na(img_height))

pw <- ggplot(dims_df, aes(x = img_width)) +
  geom_histogram(bins = 20, fill = "#4575b4", color = "white") +
  labs(title = "Image Width Distribution", x = "Width (px)", y = "Count") +
  theme_minimal(base_size = 11)

ph <- ggplot(dims_df, aes(x = img_height)) +
  geom_histogram(bins = 20, fill = "#74add1", color = "white") +
  labs(title = "Image Height Distribution", x = "Height (px)", y = "Count") +
  theme_minimal(base_size = 11)

pw + ph
```



```r
dims_df %>%
  summarise(
    unique_resolutions = n_distinct(paste(img_width, img_height)),
    most_common_width  = as.integer(names(sort(table(img_width),  decreasing = TRUE)[1])),
    most_common_height = as.integer(names(sort(table(img_height), decreasing = TRUE)[1])),
    min_width = min(img_width),  max_width = max(img_width),
    min_height = min(img_height), max_height = max(img_height)
  )
```

```
##   unique_resolutions most_common_width most_common_height min_width max_width
## 1                  1              2048               1024      2048      2048
##   min_height max_height
## 1       1024       1024
```

---

10 Missing / Data Quality Check

```r
annotations %>%
  summarise(
    pct_missing_label    = mean(is.na(label)),
    pct_missing_bbox_w   = mean(is.na(bbox_w)),
    pct_missing_bbox_h   = mean(is.na(bbox_h)),
    pct_missing_img_dims = mean(is.na(img_width) | is.na(img_height))
  ) %>%
  pivot_longer(everything(), names_to = "Field", values_to = "Pct_Missing") %>%
  mutate(Pct_Missing = percent(Pct_Missing, accuracy = 0.01))
```

```
## # A tibble: 4 x 2
##   Field                Pct_Missing
##   <chr>                <chr>
## 1 pct_missing_label    0.62%
## 2 pct_missing_bbox_w   0.62%
## 3 pct_missing_bbox_h   0.62%
## 4 pct_missing_img_dims 0.00%
```

---

11 EDA Summary Table

```r
tibble(
  Metric = c(
    "Total annotation files (images)",
    "Train images",
    "Validation images",
    "Cities represented",
```

```r
    "Total person annotations (non-ignore)",
    "Total ignore-region annotations",
    "Images with >1 pedestrian",
    "Images without pedestrian",
    "Pedestrian presence rate",
    "Mean persons per image (all)",
    "Mean persons per image (ped only)",
    "Class imbalance ratio"
  ),
  Value = c(
    comma(n_distinct(annotations$file)),
    comma(sum(images_df$split == "train")),
    comma(sum(images_df$split == "val")),
    as.character(n_distinct(annotations$city)),
    comma(sum(annotations$is_person, na.rm = TRUE)),
    comma(sum(annotations$is_ignore, na.rm = TRUE)),
    comma(sum(image_level$has_pedestrian)),
    comma(sum(!image_level$has_pedestrian)),
    percent(mean(image_level$has_pedestrian), accuracy = 0.1),
    as.character(round(mean(image_level$n_persons), 2)),
    as.character(round(mean(image_level$n_persons[image_level$has_pedestrian]), 2)),
    as.character(round(ratio, 2))
  )
)
```

```
## # A tibble: 12 x 2
##    Metric                                Value
##    <chr>                                 <chr>
##  1 Total annotation files (images)       3,475
##  2 Train images                          2,975
##  3 Validation images                     500
##  4 Cities represented                    21
##  5 Total person annotations (non-ignore) 25,166
##  6 Total ignore-region annotations       8,399
##  7 Images with >1 pedestrian             2,941
##  8 Images without pedestrian             534
##  9 Pedestrian presence rate              84.6%
## 10 Mean persons per image (all)          7.24
## 11 Mean persons per image (ped only)     8.56
## 12 Class imbalance ratio                 5.51
```

Data: Cityscapes Dataset — gtBboxCityPersons annotations (Cordts et al., 2016)